Wesley Kwiecinski

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SUMMARY

Master's Graduate in Computer Science and research software engineer focused on HPC. Has hands-on experience at the Argonne Leadership Computing Facility (ALCF) building data pipelines and visualization tooling for exa-scale scientific and AI training datasets, and M.S. thesis research characterizing I/O performance of scientific workflows. My background spans processing, and visualizing large scientific datasets in Python, deploying and debugging scientific software across diverse HPC and container environments, extending existing scientific tools, and collaborating in multi-institutional teams at Argonne, UIC, and CERN.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Rust

Data & Storage: Darshan I/O profiling, parallel/HPC filesystems, SQL / SQLite, MariaDB, JSON, NumPy, Pandas, custom clustering pipelines

HPC Systems & Deployment: Aurora, Polaris, Crux, Perlmutter; PBS job scheduling, MPI; Apptainer, Shifter, Podman container runtimes; Linux systems

Data Visualization & ML: Plotly, Matplotlib, PyTorch, Torchtitan, Scikit, EZPZ

Practices & Tools: Git/GitHub, code review, unit testing (pytest), AI-assisted development tools (Copilot, Claude code)

RESEARCH & PROFESSIONAL EXPERIENCE

Argonne Leadership Computing Facility (ALCF), Argonne National Laboratory *Summer 2026*

Research Aide

- Engineered and debugged a python-based data processing and clustering pipeline for HPC systems (MPI, MiniBatchKMeans, sparse category-probability and dense scalar features, custom JSON schema) for processing documents for AI training.
- Used AI-assisted development tools to build a visualization tool for generating a dashboard for dataset clusters and document-similarity relationships using python and matplotlib, focused on interactive-scale performance.
- Designed, tested, and debugged an LLM pre-training pipeline on Aurora and Polaris using torchtitan and ezpz to evaluate the impact of different token orderings on training and downstream task performance.

Electronic Visualization Laboratory (EVL), University of Illinois Chicago *Fall 2024 – Summer 2026*

Research Aide / M.S. Thesis Research

- Extended existing performance monitoring tools with I/O monitoring, characterization, and visualization solutions for ATLAS workflows on Perlmutter and CERN nodes and container runtimes (Apptainer, Shifter, Podman-hpc), using Python and Darshan trace analysis, in collaboration with Argonne National Laboratory and CERN.
- Integrated Darshan I/O profiling into ATLAS Athena data-processing workflows, resolving container-runtime issues to capture I/O behavior of production scientific workflows.
- Diagnosed and resolved container-runtime data-access issues on HPC systems (overlay-filesystem permission errors, shared memory exhaustion) that were blocking reliable large-scale scientific data I/O.
- Conducted research as part of the Chicagoland Computational Traineeship in High Energy Particle Physics (C^2-The-P^2) fellowship; co-authored a CHEP 2026 conference presentation and paper on the resulting I/O access-pattern findings.

Argonne Leadership Computing Facility (ALCF), Argonne National Laboratory *Summer 2025*

Research Aide

- Extended and tested performance-monitoring systems for ATLAS HEP workflows on ALCF systems, adding I/O performance monitoring support used to characterize scientific data access at scale through the C^2-The-P^2 fellowship.

Math and Computer Science Division (MCS), Argonne National Laboratory *Feb 2023 – Aug 2024*

Post-Baccalaureate Appointee (Software Engineering)

- Collaborated with Flash-X developers to build code-generation tools supporting the Flash-X multi-physics framework through an agile development process, under the supervision of senior engineers.
- Participated in Python code reviews and developed unit tests using pytest to support reliable, maintainable scientific software.
- Developed and tested automated tools to generate Fortran code for GPU data transfer and built visualization tools to analyze large-scale data generated on the Aurora supercomputer using python and matplotlib.

DDILab, Northern Illinois University *Aug 2021 – Dec 2022*

Undergraduate Research Aide

- Co-developed a web-based tool for visualizing crosswalk image data, exposing collected sensor data to lab researchers through an interactive interface, iterating through feedback.
- Improved existing crosswalk-detection code for edge deployment on Sage nodes using TensorFlow and PyTorch.
- Designed, implemented, and tested a MariaDB database system for storing and retrieving edge-collected data using SQL.

PUBLICATIONS, POSTERS & TALKS

- Kwiecinski, W. "Analysis of ATLAS Derivation I/O on High-Performance Computing Systems," Master's Thesis, 2026.
- Yildiz, O., Gartung, P., Wang, R., Kwiecinski, W. "Characterizing Event I/O Access Patterns in HEP Workflows Using Darshan," Presentation, 28th Conference on Computing in High Energy and Nuclear Physics, 2026.
- Lee, Y., Weide, K., Kwiecinski, W., O'Neal, J., Rudi, J., Dubey, A. "ORCHA – A Performance Portability System for Post Exascale Systems," 2025. https://doi.org/10.48550/arXiv.2507.09337
- Kwiecinski, W., Papka, M. "Evaluating I/O Behavior of ATLAS Derivations Across Container Runtimes on HPC Platforms," Poster, Greater Chicago Area Systems Research Workshop, 2026.

EDUCATION

M.S. in Computer Science — University of Illinois Chicago, Aug 2026

Thesis: Analysis of ATLAS Derivation I/O on High-Performance Computing Systems — 4.0 GPA

B.S. in Computer Science, emphasis in Software Engineering — Northern Illinois University, Dec 2022

Minor in General Mathematics — Magna Cum Laude

FELLOWSHIPS

Chicagoland Computational Traineeship in High Energy Particle Physics (C^2-The-P^2), 2024 – 2026

Collaboration with Argonne National Laboratory and Fermilab to advance computational methods in particle physics — c2thep2.org